Replicating Manifold Steering: Geometry of LLM Representations and Behavior

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Code, checkpoints, and figures: `https://github.com/kirillacharya/manifold-steering-replication`

Background: do LLM representations and behavior share a geometry?

Large language models build up internal representations layer by layer: at layer L, each token maps to a hidden state $h^{(L)} \in \mathbb{R}^d$, and structured concept domains may organize into geometric patterns. Hidden states for *days of the week* or *months of the year* may trace a smooth, low-dimensional manifold $\mathcal{M}_h \subset \mathbb{R}^d$ — closed loops for cyclic domains, open curves for sequential ones. Wurgaft et al. [1] ask whether this geometry *predicts behavior*: output probability vectors, embedded via the Hellinger map $p \mapsto \sqrt{p}$ (which makes the Hellinger distance $d_H(p,q) = \frac{1}{\sqrt{2}}\|\sqrt{p} - \sqrt{q}\|$ Euclidean [2]), may trace a matching *behavior manifold* $\mathcal{M}_y$. If $\mathcal{M}_h \approx \mathcal{M}_y$ isometrically, steering *along* $\mathcal{M}_h$ should produce natural behavioral transitions, while linear steering cuts off-manifold and "teleports" probability mass. The two strategies interpolate between a source concept $\hat{h}_{c_s}$ and a target $\hat{h}_{c_e}$:

$$\pi_{\text{lin}}(\alpha) = (1-\alpha)\,\hat{h}_{c_s} + \alpha\,\hat{h}_{c_e}, \qquad \pi_{\text{mfd}}(\alpha) = s\big((1-\alpha)\,u_{c_s} + \alpha\,u_{c_e}\big), \quad u_{c_i} = s^{-1}(\hat{h}_{c_i}),$$

where $s : \mathbb{R}^k \to \mathbb{R}^d$ maps intrinsic coordinates back to activation space. We replicate the paper's central figure on a small open model, then extend it to a 2D concept structure (a day×hour torus).

The experiment, step by step

Setup. All experiments use instruction-tuned **Gemma-2-2B** ($d = 2304$), patching layer 24 of 26. Two cyclic concept domains: *weekdays* (7 classes) and *months* (12 classes), probed with arithmetic prompts — "`Q: What day is two days after Monday?  A:`" (answer: *Wednesday*), 6 prompts per concept (`scripts/run_weekdays.py`, `run_months.py`). For each prompt we record the last-token hidden state $h^{(24)}$ and the output distribution over concept tokens, then

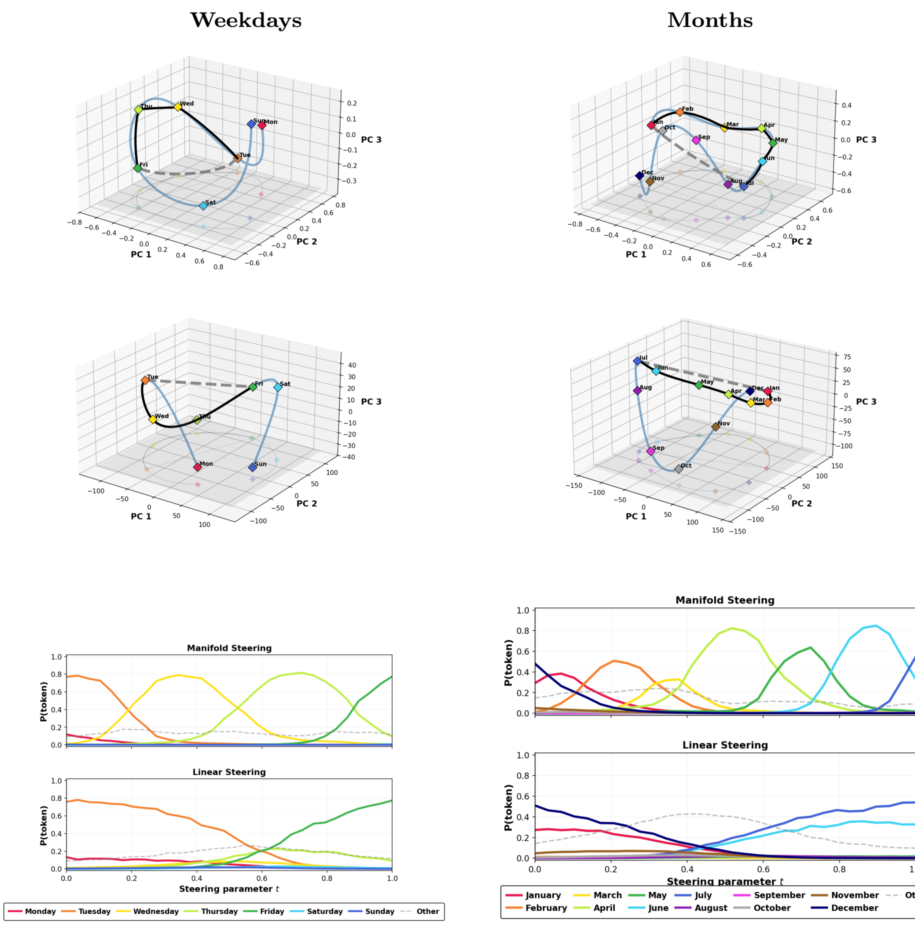

Figure 1: Gemma-2-2B layer 24, steering Tue→Fri (weekdays) and Jan→Jul (months). Rows 1–2: fitted behavior manifolds $\mathcal{M}_y$ (Hellinger–PCA) and activation manifolds $\mathcal{M}_h$ (activation–PCA) with the linear path (dashed) and manifold path (black). Row 3: concept-token probabilities along each path.

average per concept: $\hat{h}_c = \frac{1}{N_c}\sum_{i\in S_c} h_i^{(L)}$ and $\hat{p}_c = \frac{1}{N_c}\sum_{i\in S_c} p_i$.

Manifolds. A "manifold" here is simply a smooth curve through the concept centroids. The 7 weekday centroids $\hat{h}_{c_k} \in \mathbb{R}^{2304}$ span at most a 6-dimensional subspace, so PCA to r=6 (r=11 for 12 months) is lossless — it just re-expresses each centroid in coordinates $z_k \in \mathbb{R}^r$ adapted to the data. We then fit a cubic spline $\gamma : \mathbb{R} \to \mathbb{R}^r$ through the points *in concept order*, with the concept index as curve parameter: $\gamma(k) = z_k$ exactly at each knot, and between knots $\gamma(t) = a_k + b_k(t-k) + c_k(t-k)^2 + d_k(t-k)^3$, coefficients solved jointly so the curve is C^2-smooth. This gives the activation manifold $\mathcal{M}_h$: a single scalar coordinate ("where between Monday and Sunday") decodes to a full hidden state via $\mathrm{PCA}^{-1}(\gamma(t))$. The behavior manifold $\mathcal{M}_y$ is built identically from the probability centroids, except each $\hat{p}_{c_k}$ is first mapped to $\sqrt{\hat{p}_{c_k}}$ (Hellinger space), where Euclidean tools — PCA, splines, distances — respect the geometry of the probability simplex.

Steering. To steer from a source concept to a target (say Tuesday → Friday, spline indices i_s=1, i_e=4), we build the two paths of the Background section concretely. The *linear* path ignores the manifold: $\pi_{\mathrm{lin}}(\alpha) = (1-\alpha)\hat{h}_{c_s} + \alpha\hat{h}_{c_e}$, a straight segment between the two raw centroids in $\mathbb{R}^{2304}$. The *manifold* path instead interpolates the scalar spline coordinate and decodes: $\pi_{\mathrm{mfd}}(\alpha) = \mathrm{PCA}^{-1}\big(\gamma(i_s + \alpha\,(i_e - i_s))\big)$ — at $\alpha = \frac{1}{3}$ it passes through (a point near) the Wednesday centroid, at $\alpha = \frac{2}{3}$ Thursday. Both paths are sampled at 30 evenly spaced α_t. The intervention itself: the model reads a fixed, neutral base prompt (one whose correct answer is the midpoint concept), a forward hook on decoder block 24 overwrites the last-token hidden state with the path point $\pi(\alpha_t)$, the remaining two blocks and the unembedding run as usual, and we record the resulting probability of every concept token plus an aggregated "other" (`scripts/run_steering_probs.py`). If behavior follows geometry, the manifold path should hand probability mass smoothly from concept to concept while the linear path should not. Everything is deterministic; all figures regenerate from committed checkpoints without a GPU (see the repo README).

Replication of the main figure

Figure 1 replicates the paper's main result. In both domains the concept centroids trace closed, ordered loops in activation space and in behavior space — the cyclic structure of weeks and months is visibly encoded. The manifold path stays on the fitted curve and, in the probability trajectories (row 3), produces *smooth, ordered* transitions: steering from Tuesday to Friday passes probability mass through Wednesday and Thursday in sequence. Linear steering instead leaves the manifold and *teleports*: intermediate concepts never rise in order, and off-concept mass appears mid-path. One practical lesson: patch deep enough — at layer 12, keys/values from the original prompt override the patch over the 14 remaining blocks and both methods flatline; layer 24 (92% depth) is clean.

Extension: torus steering (day × hour)

Do LLMs encode *two-dimensional* concept structure? A weekday lives on one circle; a (day, hour) pair lives on a *product* of two circles — a torus. We collect layer-24 activations for all 168 prompts "`It is HH:00 on {Day}`" (7 days × 24 hours) and assign each prompt two angles, $\theta_{\mathrm{day}} = 2\pi k/7$ and $\theta_{\mathrm{hour}} = 2\pi h/24$. Each circle's 2D subspace $V_{\mathrm{day}}, V_{\mathrm{hour}} \in \mathbb{R}^{2\times d}$ is recovered by regressing centered activations onto the standardized labels $[\sin\theta, \cos\theta]$ — each label component yields one direction $d \propto \sum_i (h_i - \bar{h})\,\ell_i$ — followed by QR-orthogonalization of the two directions (`scripts/run_hierarchical_steering.py`). The two recovered planes are nearly orthogonal to each other (max $|\cos| \approx 0.18$), so day and hour occupy largely independent planes of the residual stream. Projecting each activation onto both planes gives two recovered angles, rendered on a torus — day on the major circle, hour on the minor (Fig. 2, left). The 7 day-arcs are cleanly ordered; the hour circle does not fully close, so hour is only partially circular here.

Steering in *product coordinates* respects this factorization: starting from the source activation h_{base} (Tue, 09:00), each step moves the day and hour coordinates *independently within their own planes*, $h(\alpha) = h_{\mathrm{base}} + \Delta_{\mathrm{day}}(\alpha)\,V_{\mathrm{day}} + \Delta_{\mathrm{hour}}(\alpha)\,V_{\mathrm{hour}}$, where $\Delta_c(\alpha)$ interpolates the source→target offset within each plane — the 2D analogue of walking along the manifold. Each patched activation is decoded into a day×hour probability grid: day from token probabilities,

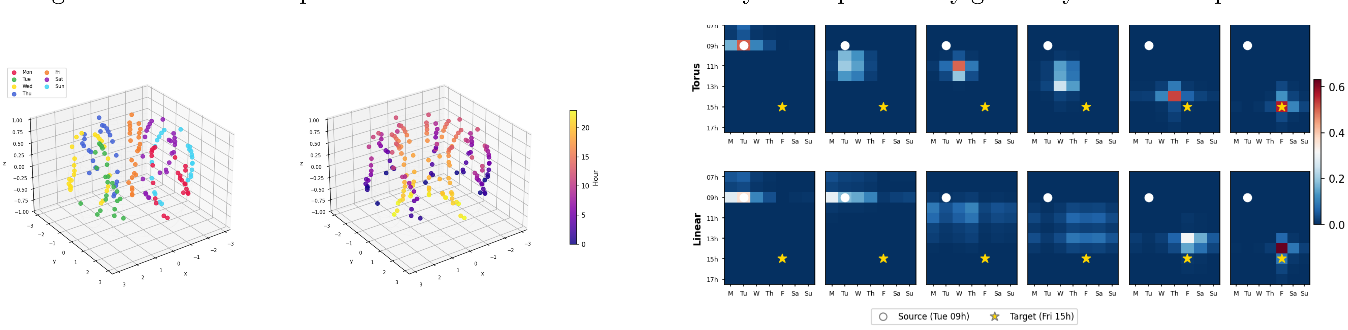

Figure 2: Left: layer-24 activations on the day×hour torus, colored by day (7 ordered arcs) and by hour. **Right:** day×hour probability grids across 6 steering steps, (Tue, 09:00) → (Fri, 15:00); top row torus (product-coordinate) steering, bottom row linear steering.

hour by nearest centroid in the V_{hour} plane (two-digit hours are multi-token for Gemma, so subspace readout is more faithful). Steering $(\text{Tue}, 09{:}00) \rightarrow (\text{Fri}, 15{:}00)$ over six steps (Fig. 2, right), the bright cell walks diagonally — both coordinates move smoothly — while linear steering moves the day correctly but teleports the hour, jumping to 15:00 only at the final step: the same off-manifold failure mode, now visible per coordinate.

Takeaway. Representation geometry is not decoration: moving along it produces natural behavior, and ignoring it breaks transitions exactly as the manifold picture predicts. Code, checkpoints, and full reproduction instructions: https://github.com/kirillacharya/manifold-steering-replication.

References

[1] D. Wurgaft, C. Rager, M. Kowal, V. Shyam, S. Feucht, U. Bhalla, T. Haklay, E. Bigelow, R. Sarfati, T. McGrath, O. Lewis, J. Merullo, N. D. Goodman, T. Fel, A. Geiger, and E. S. Lubana. Manifold steering reveals the shared geometry of neural network representation and behavior. *arXiv preprint arXiv:2605.05115*, 2026.

[2] S. Amari and H. Nagaoka. *Methods of Information Geometry.* American Mathematical Society, 2000.